

Tema 7

Preguntas Tipler

Potencial en Campo Uniforme

Ejercicio 20

$$V = 500 \text{ V} \quad d = 0,1 \text{ m}$$

$$a) E = \frac{V}{d} = 5000 \text{ N/C}$$

b) La placa positiva tiene mayor potencial

$$c) W = \int_a^b \vec{F} \cdot d\vec{s} = F \cdot d = q \cdot E \cdot d$$

$$W = 1,6 \cdot 10^{-19} \cdot 5000 \cdot 0,1 = 8 \cdot 10^{-17} \text{ J}$$

$$d) W = -\Delta U = -8 \cdot 10^{-17} \text{ J}$$

$$e) K_0 + U_0 = K_3 + U_3$$

$$\frac{1}{2} m \cdot v^2 = 8 \cdot 10^{-17} \rightarrow v^2 = 1,76 \cdot 10^{-14} \rightarrow v = 1,33 \cdot 10^7 \text{ m/s}$$

Problema 21

$$E = 2000 \text{ V/C} \quad d = 3 \text{ m/C}$$

$$a) V(4) - V(0) = - \int_0^4 E \cdot dx = -2000 \cdot 4 = -8000 \text{ V}$$

$$b) \Delta U = q \cdot \Delta V = -8000 \cdot 3 \cdot 10^{-6} = -0,024 \text{ J}$$

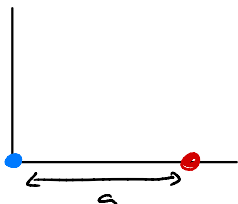
$$K_0 + U_0 = K_4 + U_4 \rightarrow K_4 = -U_4 = 0,024 \text{ J}$$

$$d) V(x) = -E \cdot x$$

Potencial para Partículas Puntuales

Problema 32

$$a) V(x) = k \cdot e \cdot \left(\frac{3}{x} + \frac{-2}{x-a} \right)$$



$$b) V(x) = 0 \rightarrow k \cdot e \cdot \left(\frac{3}{x} - \frac{2}{x-a} \right) = 0$$

$$\frac{3}{x} = \frac{2}{x-a} \rightarrow 3x - 3a = 2x \rightarrow x = 3a$$

$$c) E(x) = 0 \rightarrow \frac{3}{x^2} = \frac{2}{(x-a)^2}$$

$$3x^2 - 6x + 3a^2 = 2x^2$$

$$x^2 - 6xa + 3a^2 = 0$$

$$(x-3a)^2 - 9a^2 + 3a^2 = 0$$

$$(x-3a)^2 = 6a^2$$

$$x-3a = \sqrt{6}a \rightarrow x = 3a \pm \sqrt{6}a$$

$$x = (3 \pm \sqrt{6})a$$

$$d) W = q \cdot \Delta V$$

$$V(a/2) \cdot k \cdot e \cdot \left(\frac{3}{a/2} + \frac{0.2}{a/2} \right) = k \cdot e \cdot \frac{6.4}{a}$$

$$W = e \cdot (V(x) - V(a/2)) = k \cdot e^2 \cdot \left(\frac{3}{x} - \frac{2}{(x-a)} - \frac{6.4}{a} \right)$$

Determinación del Campo Eléctrico en función de la Potencial

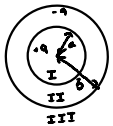
Problema 35

$$a) V_b - V_a = - \int_a^b E \cdot dl \rightarrow E < 0 \rightarrow V_b - V_a > 0$$

$$b) V_b - V_a = - \int_a^b E \cdot dl$$

$$-1 \cdot 10^5 = E (6 \cdot 2) \rightarrow E = -25000 \text{ V/m}$$

Problema 45



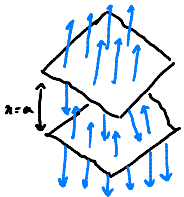
$$\oint E \cdot dA = \frac{q_{enc}}{\epsilon_0}$$

$$E \cdot 2\pi r L = \frac{\lambda L}{\epsilon_0} \rightarrow E(r) = \frac{\lambda}{2\pi \epsilon_0 r}$$

$$V_a - V_b = \int_a^b E(r) dr \rightarrow V_a - V_b = \int_a^b \frac{\lambda}{2\pi \epsilon_0 r} dr$$

$$V_a - V_b = \frac{\lambda}{2\pi \epsilon_0} [\ln(r)]_a^b \rightarrow V_a - V_b = \frac{\lambda}{2\pi \epsilon_0} \cdot \ln\left(\frac{b}{a}\right) = \frac{q}{2\pi \epsilon_0 L} \cdot \ln\left(\frac{b}{a}\right)$$

Problema 49



a) Región 1: $x \leq 0$

$$\oint E \cdot dA = \frac{q_{enc}}{\epsilon_0} \rightarrow E \cdot A = \frac{\sigma A}{\epsilon_0} \rightarrow E = \frac{\sigma}{\epsilon_0}$$

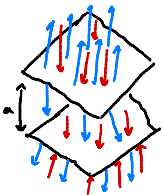
$$V_x - V_0 = \int_{-x}^0 \frac{\sigma}{\epsilon_0} dx = - \frac{\sigma}{\epsilon_0} (-x) = \frac{\sigma}{\epsilon_0} x$$

Región 2: $0 < x < a \rightarrow E = 0 \rightarrow V = 0 \text{ V}$

Región 3: $x \geq a \rightarrow E = -\frac{\sigma}{\epsilon_0}$

$$V_a - V_x = \int_a^x -\frac{\sigma}{\epsilon_0} dx = -\frac{\sigma}{\epsilon_0} (x - a)$$

b)



Región 1: $x \leq 0: E = 0 \rightarrow V = 0$

Región 2: $E = -\frac{\sigma}{\epsilon_0} \quad 0 < x < a$

$$V_0 - V_b = -\frac{\sigma}{\epsilon_0} x$$